The ACS system is a mixed system, sharing the biomass boiler with the heating system and using an accumulation tank.

1. ACS Demand Calculation (`Demanda de ACS`)

This calculates the daily volume of hot water needed, including distribution losses.

Formula: `Daily ACS Demand (l/day) = Number of Occupants × Daily Needs per Person (l/p·day)`

Where:

- **Number of Occupants:** Based on DB HE Annex F, Table a (Minimum occupancy for private residential use).
- **Daily Needs per Person:** Fixed at **28 l/p·day** (at 60°C).

Calculation from Pages 19 & 47:

Option 2: 2 Bedrooms, 3 Occupants

- **Base Demand:** `3 occupants × 28 l/p·day = 84 l/day`
- **Distribution Losses (estimated at 5%):** `84 l/day × 0.05 = 4.2 l/day`
- **Total Daily Demand (excluding tank losses):** `84 + 4.2 = 88.2 l/day`

Option 3: 3 Bedrooms, 4 Occupants

- **Base Demand:** `4 occupants × 28 l/p·day = 112 l/day`
- **Distribution Losses (estimated at 5%):** `112 l/day × 0.05 = 5.6 l/day`
- **Total Daily Demand (excluding tank losses):** `112 + 5.6 = 117.6 l/day`

2. Heat Loss Calculation for the Accumulation Tank (`Pérdidas en depósito`)

This is a critical calculation to determine the total energy demand, as it includes losses from the hot water storage tank.

Formula (Page 48): `Q = A · U · ΔT · Number of hours in the period`

Simplified using the Tank Loss Coefficient: `Monthly Losses (Wh) = Loss Coefficient (W/°C) × ΔT (°C) × Hours in Month`

Where:

- **Loss Coefficient (A·U):** Provided in the system specs.
  - Options 1 & 2 (100L tank): `0.5 W/°C`
  - Option 3 (150L tank): `0.8 W/°C`
- **ΔT:** Temperature difference between tank interior (65°C) and ambient (20°C) = **45°C**

Calculation for Option 2 (Page 49):

- **For January (744 hours):** `Q_jan = 0.5 W/°C × 45°C × 744 h = 16,740 Wh = 16.74 kWh`
- **Total Annual Tank Losses:** `197.10 kWh/year`
- **Average Daily Tank Losses:** `197,100 Wh/year / 365 days = 540 Wh/day`

3. Equivalent Daily Volume of Tank Losses

This converts the energy lost from the tank back into an equivalent volume of water that would need to be heated.

Formula (Page 49): `Equivalent Volume (l/day) = Daily Energy Loss (Wh/day) / [ΔT × C_a × ρ_a]`

Where:

- **ΔT:** Temperature rise (60°C - T_cold). `T_cold` is the annual average cold water temperature at the project altitude.
- **C_a:** Specific heat capacity of water = **1.163 Wh/(kg·K)**
- **ρ_a:** Density of water ≈ **1 kg/l**

Calculation for Option 2 (Page 49):

- Daily Energy Loss = `540 Wh/day`
- ΔT = `60°C - 9.74°C = 50.26°C` (9.74°C is the average cold water temp for the location)
- **Equivalent Volume =** `540 Wh/day / (50.26 °C × 1.163 Wh/(kg·K) × 1 kg/l) ≈ 9.2 l/day`

Total Demand for HE4 Applicability Check (Option 2):

- `88.2 l/day + 9.2 l/day = 97.4 l/day`
- Since this is **below 100 l/day**, Option 2 is **not subject** to the HE4 renewable contribution requirement.

4. Total Annual Energy Demand for ACS (`Demanda energética anual`)

This calculates the total energy required to heat the water and compensate for all losses over a year.

Formula (Page 51): `Monthly ACS Energy (kWh) = V_ACS_month (l) × C_a × ρ_a × (60°C - T_cold_month) / 1000`

Where:

- `V_ACS_month` is the monthly volume of water used (at 60°C).
- `T_cold_month` is the monthly average cold water temperature.

Calculation for Option 3 (Page 51):

- The document provides a detailed month-by-month calculation.
- **Total Annual Energy Demand (including tank losses):** `2,823.00 kWh/year`
- **Useful Floor Area (Option 3):** `128 m²`
- **Specific Demand:** `2,823.00 kWh/year / 128 m² = 22.05 kWh/m²·year`

5. Renewable Energy Contribution Calculation (`Contribución renovable`)

This proves compliance with HE4, showing that over 60% of the ACS demand is met by renewable energy (biomass).

Step-by-Step Calculation from Pages 52 & 53:

Step 1: Final Energy Consumption

- **Formula:** `Final Energy (kWh/m²·year) = ACS Demand (kWh/m²·year) / Boiler Efficiency`
- **Calculation:** `22.05 kWh/m²·year / 0.93 = 23.71 kWh/m²·year`

Step 2: Calculate Renewable Portion of Final Energy

- **Formula:** `Renewable Final Energy = Final Energy × (f_ep,ren / f_ep,tot)`
- **Factors for Biomass Pellets (Page 57):**
    - `f_ep,ren` = 1.028
    - `f_ep,tot` = 1.113

- **Ratio:** `1.028 / 1.113 = 0.9236`
- **Calculation:** `23.71 kWh/m²·year × 0.9236 = 21.90 kWh/m²·year`

Step 3: Convert Back to Renewable Demand

- **Formula:** `Renewable Demand = Renewable Final Energy × Boiler Efficiency`
- **Calculation:** `21.90 kWh/m²·year × 0.93 = 20.37 kWh/m²·year`

Step 4: Calculate Renewable Percentage

- **Formula:** `% Renewable = (Renewable Demand / Total ACS Demand) × 100`
- **Calculation:** `(20.37 / 22.05) × 100 = 92.4%`

Compliance Check (HE4):

- **Requirement for demand < 5000 l/day:** Minimum **60%** renewable contribution.
- **Result:** `92.4% > 60%` ✅ **COMPLIES**

Summary of ACS System Data:

Aspect	Option 2	Option 3
Occupants / Bedrooms	3 / 2	4 / 3
Daily Demand (w/ distribution losses)	88.2 l/day	117.6 l/day
Accumulator Volume	100 L	150 L
Tank Loss Coefficient (A·U)	0.5 W/°C	0.8 W/°C
Total Demand (w/ tank losses)	97.4 l/day	(N/A, calculated in kWh)
HE4 Applicable?	**No** (<100 l/day)	**Yes** (>100 l/day)
Specific Annual Demand	(Not fully calculated)	**22.05 kWh/m²·year**
Renewable Contribution	(Not required)	**92.4%**

The document provides a complete traceability from the basic water consumption needs to the final proof that the system, powered by a biomass boiler, comfortably exceeds the legal requirements for renewable energy use in domestic hot water production.